

Chest CT Angiography (CTA)

ACG: A-0471 (AC)

[Link to Codes](#)

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- Clinical Indications
- Evidence Summary
 - Background
 - Criteria
 - Rationale
 - Related CMS Coverage Guidance
- References
- Footnotes
- Codes

Clinical Indications

- Chest CT angiography (CTA) may be indicated for **1 or more** of the following:
 - ☐ Anatomy or structural vascular defect evaluation needed, as indicated by **1 or more** of the following(3)(4)(5):[N](#)
 - Arterial pseudoaneurysm(7)
 - Follow-up after surgical correction of vascular anomaly(8)(9)
 - Pulmonary arteriovenous malformation(10)(11)
 - Vascular rings(6)
 - ☐ Aortitis or large vessel vasculitis, known or suspected, as indicated by **1 or more** of the following(12)(13)(14)(15):[N](#)
 - Infectious vasculitis (eg, syphilis, mycotic aneurysm)
 - Inflammatory vasculitis (eg, Takayasu arteritis, giant cell arteritis, systemic lupus erythematosus)(20)(21)
 - ☐ Central vein evaluation needed, as indicated by **1 or more** of the following(3)(6):[N](#)
 - Facial or upper extremity swelling
 - Potential access site for hemodialysis
 - Superior vena cava syndrome(22)
 - Chronic thromboembolic pulmonary hypertension, suspected(23)(24)(25)(26)(27)(28)(29)[N](#)
 - Congenital malformation of chest, lungs, mediastinum, or great vessels, known or suspected, when additional imaging required for management(10)(15)(37)(38)(39)(40)(41)[N](#)
 - Hemoptysis (ie, massive, nonmassive, or recurrent hemoptysis), and need for preprocedural planning(44)(45)[N](#)
 - Medium vessel vasculitis, suspected (eg, vasculitis affecting main visceral arteries)(46)[N](#)
 - Postoperative or postprocedural evaluation needed to assess luminal patency or complications(47)[N](#)
 - ☐ Preoperative or preprocedural planning needed, as indicated by **1 or more** of the following:[N](#)
 - Aortic valve disease, and transcatheter aortic valve implantation or replacement planned(48)(49)(50)(53)(54)
 - Chronic thromboembolic pulmonary hypertension, and balloon pulmonary angioplasty planned(26)(29)(51)(55)
 - Chronic thromboembolic pulmonary hypertension, and pulmonary thromboendarterectomy planned(27)(29)(52)(55)
 - ☐ Pulmonary embolism evaluation needed, as indicated by **1 or more** of the following(56)(57)(58)(59)(60)(61)(62)(63)(64):[N](#)
 - High clinical probability of pulmonary embolism
 - Low or moderate clinical probability of pulmonary embolism and abnormal results of D-dimer test(80)
 - Recurrent or residual pulmonary embolism, known or suspected(55)
 - ☐ Thoracic aortic aneurysm or dissection evaluation needed, as indicated by **1 or more** of the following(13)(15)(41)(82)(83)(84):[N](#)
 - Aortic regurgitation murmur

- Bicuspid aortic valve, known[A](5)(91)
- Blunt chest trauma, penetrating aortic trauma, or iatrogenic trauma from aortic instrumentation(92)(93)
- Chest or upper back pain, unexplained, with acute or subacute onset
- Dysphagia
- First-degree relative of patient with thoracic aortic aneurysm and/or dissection(91)
- Genetic predisposition to aortic disease (eg, Loeys-Dietz syndrome, confirmed genetic mutations known to predispose to aortic aneurysms and aortic dissection, vascular Ehlers-Danlos syndrome, Turner syndrome, Marfan syndrome), with imaging at initial diagnosis, regular surveillance based on specific condition, preconception planning, pregnancy monitoring, or postpartum evaluation(91)(94)
- Hoarseness, with suspected recurrent laryngeal nerve palsy
- Inflammatory vasculitis (eg, giant cell arteritis, Takayasu arteritis, Behcet disease), known or suspected(95)(96)(97)(98)
- Intramural hematoma, known or suspected(99)
- Penetrating atherosclerotic ulcer, known or suspected(99)
- Plain chest x-ray abnormal (eg, widened mediastinum, enlarged aortic knob, tracheal displacement)
- Postoperative monitoring of thoracic aortic surgery(88)(100)(101)
- Preoperative planning for thoracic aortic repair (ie, assessment of spinal cord circulation via great anterior radiculomedullary artery)(100)(101)
- Thoracic aortic aneurysm or dilatation, known or suspected, as indicated by **1 or more** of the following(91)(100)(101):
 - Abnormal finding on chest x-ray or echocardiography suggestive of thoracic aortic disease
 - New or worsening signs or symptoms(102)
 - Surveillance after surgical repair, at 1, 3, 6, and 12 months
 - Surveillance of stable patient without thoracic aorta imaging in past 6 months
- Thoracic aortic dissection, known or suspected, as indicated by **1 or more** of the following(91)(100)(101):
 - Abnormal finding on chest x-ray or echocardiography suggestive of thoracic aortic dissection(62)
 - New or worsening signs or symptoms(62)(102)
 - Preoperative or preprocedural planning needed
 - Surveillance after initial dissection or surgical repair, at 1, 3, 6, and 12 months post dissection
 - Surveillance of stable patient without thoracic aorta imaging in past year
- Transthoracic echocardiogram findings abnormal(91)
- Trauma of chest, known or suspected(103)(104)(105)(106)(107)(108)(109)■
- ☐ Repeat evaluation of specific area or structure with same imaging modality, and **ALL** of the following:
 - Clinical need for repeat imaging, as indicated by **1 or more** of the following:
 - Change in clinical status (eg, worsening symptoms or new associated symptoms), and findings may impact treatment
 - Need for interval reassessment that may impact treatment plan
 - Need for re-imaging either prior to or after performance of invasive procedure
 - Prior imaging results of specific area or structure with same imaging modality documented and available for comparison

Evidence Summary

Background

CTA consists of CT scanning after intravenous injection of contrast material to better characterize vascular anatomy and pathology. Unlike standard invasive x-ray and digital subtraction angiography, it is noninvasive and rapidly provides information about extravascular and perivascular anatomy. CTA multislice helical scanning with 3-dimensional reconstruction software can provide detailed, visually familiar views of complex anatomy. However, the procedure involves exposure to radiation and contrast material, with risk of allergic reaction and nephrotoxicity.(1)(2) (**EG 2**)

Criteria

The evidence for the clinical indications found in this guideline includes 84 published peer reviewed articles, 17 specialty society or other evidence-based guidelines, and 1 Cochrane systematic review.

For anatomy or structural vascular defect evaluation, evidence demonstrates at least moderate certainty of at least moderate net benefit. (**RG A1**) CTA is a reliable and accurate modality for further delineation of abnormalities of the thoracic vasculature, including the aorta, pulmonary arteries, and pulmonary veins. In the pediatric population, CTA may be preferred over MRA or echocardiography for initial evaluation due to less patient sedation and shorter scan time, allowing for better direct patient monitoring.(3)(6) (**EG 2**) A review article notes that chest CTA or MRA should be included as part of the initial workup for thoracic arterial pseudoaneurysms.(7) (**EG 2**)

For aortitis or large vessel vasculitis, evidence demonstrates at least moderate certainty of at least moderate net benefit. (**RG A1**) CTA and MRA have largely replaced conventional angiography for diagnostic confirmation and assessment of infectious or inflammatory vasculitis.(16)(17) (**EG 2**) CTA and MRA are noninvasive and also reveal information about the vessel wall, surrounding anatomy, and lumen.(4)(18)(19) (**EG 2**)

For central vein evaluation, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Chest CTA is useful for the evaluation of central veins when central venous obstruction is suspected and for identification of potential venous access sites for hemodialysis.(3)(6) **(EG 2)**

For chronic thromboembolic pulmonary hypertension, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Chest CTA may be more sensitive than conventional pulmonary angiography for detection of central thrombotic disease,(30) and it can help to determine which patients are appropriate candidates for endarterectomy.(31) **(EG 2)** However, a specialty society guideline notes that the initial test that is recommended is a lung ventilation perfusion scan, and cautions that a normal chest CTA is not sufficient to refute the presence of operable chronic thromboembolic pulmonary hypertension, as the authors continue to regard angiography to be the gold standard for this indication.(23) **(EG 2)** For the diagnosis of chronic thromboembolic pulmonary hypertension, a systematic review and meta-analysis of 10 studies (734 patients) evaluating the efficacy of CT pulmonary angiography for the diagnosis of chronic thromboembolic pulmonary hypertension found a pooled sensitivity and specificity of 98% and 99%, respectively.(32) **(EG 1)** A review article states that lung ventilation perfusion scan has sensitivity and specificity of at least 96% and 90%, respectively. Lung ventilation perfusion scan also has a higher sensitivity than CTA for the diagnosis of distal forms of the disease.(33) **(EG 2)** Chest CTA can be used to diagnose chronic thromboembolic pulmonary hypertension in children and adolescents.(34) **(EG 2)** Chest CTA is also useful for evaluation of possible nonthrombotic causes of pulmonary hypertension.(35)(36) **(EG 2)**

For congenital thoracic malformations, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Review articles note that chest CTA, with its high spatial resolution and fast acquisition times, is useful for evaluating congenital malformations of the chest, lungs, mediastinum, or great vessels.(8)(10)(42) **(EG 2)** Another review article states that chest CTA should be considered a first-line imaging modality for evaluation of the thoracic aorta in cases of suspected congenital thoracic aortic disease.(43) **(EG 2)**

For hemoptysis (ie, massive, nonmassive, or recurrent hemoptysis), and need for preprocedural planning, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Specialty society appropriateness criteria state that chest CTA is appropriate for assessing bronchial and nonbronchial arteries prior to bronchial artery embolization.(44) **(EG 2)** Another review article states that CTA is useful for identifying abnormal pulmonary vasculature, providing a guide for surgical planning or percutaneous catheter embolization.(39) **(EG 2)**

For medium vessel vasculitis, suspected, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A specialty society guideline states that CTA is usually appropriate in the evaluation of suspected medium vessel vasculitis.(46) **(EG 2)**

For postoperative or postprocedural evaluation, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A specialty society guideline supports the use of chest CTA after endovascular repair for surveillance of endoleaks, stent migration or fracture, or increase in luminal diameter.(47) **(EG 2)**

For preoperative or preprocedural planning needed, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Practice guidelines recommend CTA of the chest, abdomen, and pelvis for assessment of access vessels (femoral, subclavian, iliac, aorta) before proceeding with transcatheter aortic valve implantation or replacement.(48)(49)(50) **(EG 2)** CTA of the chest can be performed in patients with chronic thromboembolic pulmonary hypertension before proceeding with balloon pulmonary angioplasty or pulmonary thromboendarterectomy.(51)(52) **(EG 2)**

For pulmonary embolism, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** CTA is considered the first-line diagnostic test in patients with high clinical suspicion or abnormal D-dimer test due to its extremely high negative predictive value.(58)(65)(66)(67)(68)(69) **(EG 1)** Studies have shown CTA to have sensitivity of 83% to 100% and specificity of 89% to 97%, with an overall negative predictive value approaching 99%.(58) **(EG 2)** The Prospective Investigation of Pulmonary Embolism Diagnosis II study confirmed that chest CTA is the recommended first-line imaging test in most patients, and simultaneous performance of CT venography of the legs may help to improve test sensitivity.(70) **(EG 1)** A meta-analysis of 36 studies found that the presence of right ventricular dilatation on chest CTA in patients with acute pulmonary embolism was associated with increased 30-day mortality, suggesting that CTA may be useful for risk assessment in this patient population.(71) **(EG 1)** During pregnancy, current reviews and guidelines recommend use of CTA, rather than digital subtraction angiography, when lung ventilation perfusion scanning is nondiagnostic and additional imaging evaluation is required.(72)(73)(74)(75) **(EG 1)** Because the prevalence of pulmonary embolism may be substantially lower in ambulatory children and adolescents than in adults, some authors suggest that CTA should be considered in this age group only after careful review of plain chest x-ray for other conditions that might mimic signs or symptoms of pulmonary embolism, such as consolidation or pneumothorax.(76) **(EG 2)** This is especially true for children without known major risk factors for pulmonary embolism, such as immobilization history, hypercoagulable state, excess estrogen state, indwelling central venous line, or prior history of pulmonary embolism or deep venous thrombosis.(77)(78) **(EG 2)** For pediatric patients in whom there is a high clinical suspicion for acute pulmonary embolism, review articles note that multidetector CTA, with its reduced radiation burden, can be an effective imaging technique for pulmonary embolism diagnosis in children.(79)(80)(81) **(EG 2)**

For thoracic aortic aneurysm or dissection, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** Invasive catheter-based angiography has been largely replaced by noninvasive imaging modalities, such as CTA and MRA, for evaluation of both chronic and acute thoracic aortic syndromes.(4)(17)(85) **(EG 2)** CTA has been found to significantly reduce aortic and coronary motion artifact, making it an appropriate imaging test for acute aortic emergencies, including dissection and rupture.(19)(86) **(EG 2)** CTA can provide detailed visualization of the entire aorta and its surrounding structures, and can distinguish between different aortic pathologies. CTA has been shown to have a sensitivity of up to 100% and a specificity of 98% to 99% for the detection of thoracic aortic diseases.(13)(82) **(EG 2)** A specialty society guideline indicates that CTA is appropriate for the evaluation of suspected nontraumatic thoracic aortic dissection.(87) **(EG 2)** CTA is also useful for evaluating possible postoperative complications.(88)(89)(90) **(EG 2)** Clinical guidelines recommend the performance of CTA for confirmation of suspected thoracic aortic disease in patients with abnormal aortic findings on chest x-ray, and suggest that multidetector CTA from the thoracic inlet to the common femoral arteries should be considered as the first-line diagnostic imaging modality for the evaluation of descending thoracic aortic pathology.(13) **(EG 2)**

For trauma of the chest, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** CTA is a quick and accurate means of assessing patients; it provides information about vascular integrity, and it also provides the usual comprehensive structural and anatomic information provided by standard CT scan.(104) **(EG 2)** A variety of injuries from blunt and iatrogenic trauma can be diagnosed, including incomplete or complete aortic rupture, traumatic aortic or arterial dissection, traumatic hematoma, pseudoaneurysm after endovascular repair procedures, foreign body embolism, and complications related to pulmonary artery or central venous catheters, pacemakers, or other percutaneous procedures.(92)(104)(110) **(EG 2)** Chest CTA may also assist with assessment of hemoptysis by delineation of bronchial vessel anatomy, which is often the source for hemoptysis.(111)(112)(113) **(EG 2)** Review articles and professional society guidelines report that CTA and CT scan are considered appropriate imaging methodologies to identify aortic injury after chest trauma.(105)(107)(108)(109)(114) **(EG 2)**

Rationale

Use of this MCG care guideline helps the clinician determine if a particular treatment, medication, or service might be appropriate for a specific patient, taking into account their unique health complexities.

Use of these evidence-based clinical criteria to support decision making benefits the patient by identifying patient-specific complex clinical factors and conditions, promoting personalized treatment. Utilizing evidence-based clinical criteria promotes patient safety by helping ensure that potential patient benefits outweigh the risks. In addition, the use of evidence-based guidelines can increase consistency in treatment thresholds, leading to less variation in care and promoting equitable treatment among patients.

Related CMS Coverage Guidance

This guideline supplements but does not replace, modify, or supersede existing Medicare regulations or applicable National Coverage Determinations (NCDs) or Local Coverage Determinations (LCDs).

Code of Federal Regulations (CFR): 42 CFR 419.22(115); 42 CFR 422.101(116)

Internet-Only Manual (IOM) Citations: CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 14 - Medical Devices(117); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 15 - Covered Medical and Other Health Services(118); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 16 - General Exclusions from Coverage(119)

Medicare Coverage Determinations: Medicare Coverage Database(120)

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Footnotes

[A] Patients with bicuspid aortic valve are known to have an increased likelihood of developing ascending thoracic aortic aneurysms and dissections, and warrant evaluation of the aortic root and ascending thoracic aorta.(82)(83) Bicuspid aortic valve is also known to be associated with aortic coarctation, further warranting evaluation of the thoracic aorta.(5) [A in Context Link 1]

Codes

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